

# Matthias Mifsud: Interactive Exploration Dashboard

## What You Build

A Streamlit dashboard with a per-subject viewer displaying MRI slices with detections colour-coded as true positives, false positives, and false negatives, plus a dataset summary table showing performance metrics across all 72 subjects.

## Technology

|               |                                                              |
|---------------|--------------------------------------------------------------|
| Framework     | Streamlit (launch with: streamlit run dashboard.py)          |
| Visualisation | matplotlib for slice rendering, pandas for the summary table |
| Data handling | NiBabel, NumPy, SciPy                                        |

## Files You Need

| File Type           | Location on Shared Drive                           | Naming Pattern         | Local Folder        |
|---------------------|----------------------------------------------------|------------------------|---------------------|
| T2* volumes         | nnUNet_raw/Dataset001_VALDO/imagesTr/              | VALDO_XXX_0002.nii.gz  | data/volumes/       |
| Ground-truth labels | nnUNet_raw/Dataset001_VALDO/labelsTr/              | VALDO_XXX.nii.gz       | data/labels/        |
| Predicted masks     | nnUNet_results/.../predictions_with_probabilities/ | VALDO_XXX.nii.gz       | data/predictions/   |
| Probability maps    | nnUNet_results/.../predictions_with_probabilities/ | VALDO_XXX.npz          | data/probabilities/ |
| Evaluation CSV      | nnUNet_results/.../evaluation/                     | evaluation_results.csv | data/               |

## Step 1: Set Up and Load Data

Download the shared outputs. Write a data loading module. Load the evaluation CSV into a pandas DataFrame at startup. Install: pip install streamlit nibabel numpy scipy matplotlib pandas.

## Step 2: Implement the TP / FP / FN Classification

For a given subject and threshold: apply the threshold to the probability map, run `scipy.ndimage.label` on both the prediction and ground truth to find connected components, match them by checking voxel overlap. A predicted component overlapping a ground-truth component is a true positive (green). A predicted component with no overlap is a false positive (red). A ground-truth component with no matching prediction is a false negative (yellow).

**U0001F9E0 AI Concept:** Implementing this matching yourself is how you develop a concrete understanding of what sensitivity ( $TP / (TP + FN)$ ) and precision ( $TP / (TP + FP)$ ) actually measure at the detection level.

### Step 3: Build the Per-Subject Viewer

In Streamlit: a subject dropdown (`st.selectbox`), a slice slider (`st.slider`), a threshold slider (`st.slider`, 0.0 to 1.0), and a matplotlib-rendered slice with the colour-coded overlay displayed via `st.pyplot()`. Below the image, show the metrics at the current threshold. Everything updates automatically in Streamlit when any slider changes.

### Step 4: Build the Dataset Summary Table

Display the evaluation CSV as a sortable table using `st.dataframe()`. Columns: subject ID, ground-truth count, predicted count, sensitivity, false positives, F1, Dice. This provides an at-a-glance overview of model performance across all subjects.

### Step 5: Write Documentation

README.md with setup instructions. Report should describe the classification algorithm, include screenshots, and discuss patterns in the model's performance.

### Optional Extensions

Add dataset-level charts: a histogram of microbleed counts per subject, a bar chart of per-subject F1 sorted from worst to best. Make the table clickable so selecting a row navigates to that subject's viewer.